

unhealthy gas generation and also reduce the power consumption.

## The solution

KMC identified solar-powered LED streetlights as the best fit for the purpose. The previous sodium vapour lights consumed 400 watts of energy each. LED streetlights, on the other hand, generate higher brightness level at less than half the power consumption of sodium vapour streetlights. Solar panels coupled with these LED streetlights further help to eliminate the compulsion of grid electricity.

However, majority of solar street-lighting setups come with batteries and associated challenges like occasional requirements for battery replacement and maintenance, high operational costs and risk of battery theft. Keeping all these conditions in mind, Dr Gonchaudhuri decided to go with 180W batteryless grid-connected solar LED streetlights.

## The complete setup

Before installation, the team had to ensure that the area met certain criteria for installation. First, since streetlights would be batteryless, the locality should have a stable electricity presence. Metropolitan cities in India have over 90 per cent availability of electricity, making the environment suitable for the product. Second, the site should have a reliable grid and proper poles for installation. Finally, they required permission of State Electricity Regulatory Commission for connecting a microinverter to the grid. The microinverter is required to transfer the generated solar energy to the grid. Once these arrangements were acknowledged, the team went ahead with the installation.

The total setup included 50 batteryless carbon-neutral grid-connected solar LED streetlights with the following specifications:

- 180W LED luminaire
- Total 315W (approx. 2x150W)

## THE COST AND RETURN STRUCTURE

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Month and year of implementation                          | January 2015                                                                                                                                                                                                                                                                                                                                                                                                         |
| Number of streetlights installed                          | 50                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Approx. cost of each streetlight with intelligent control | ₹ 60,000                                                                                                                                                                                                                                                                                                                                                                                                             |
| Estimated total project cost                              | ₹ 3.2 million                                                                                                                                                                                                                                                                                                                                                                                                        |
| Benefits                                                  | <ul style="list-style-type: none"> <li>• Very low utility electricity usage</li> <li>• More than 50% less luminaire energy consumption (220W less)</li> <li>• Annual 72-tonne CO<sub>2</sub> reduction</li> <li>• No operational cost</li> </ul>                                                                                                                                                                     |
| Expected ROI                                              | Within 4 years                                                                                                                                                                                                                                                                                                                                                                                                       |
| Key figures                                               | 90-day energy data: <ul style="list-style-type: none"> <li>• Generated energy exported: 5400kWh</li> <li>• Energy imported for consumption: 5300kWh</li> <li>• Excess energy generated: 100kWh</li> </ul> Monthly utility bill estimations (CESC): <ul style="list-style-type: none"> <li>• Bill of April 2014 (before installation): ₹ 30,000</li> <li>• Bill of April 2015 (after installation): ₹ 1800</li> </ul> |

Data source: Dr S.P. Gonchaudhuri and MNRE

capacity multicrystalline external solar PV panels

- Pole-top mounting frame for holding the panels
- 300W microinverter for connecting to the grid
- Intelligent controller to enable automated on/off and auto-dimming
- 9m long steel tubular poles

Dr Gonchaudhuri explains, "The intelligent controller manages LEDs during night time using the quantum of energy pushed during day. In necessity, the controller operates a dimmer circuit to minimise the consumption after midnight. However, the same cannot be felt physically."

Each streetlight including all the above-mentioned features cost ₹ 60,000. The total project cost was around ₹ 3.2 million.

## The returns

The installation completed in January 2015, and since then it has brought major benefits to the administration.

**Massive electricity savings.** The installation of 180W LEDs in place of 400W sodium vapour lamps instantly reduced the energy consumption by more than 50 per cent (400-180 = 220W savings). Furthermore, solar energy generated by panels is utilised throughout the night, cutting down the grid electricity consumption as much as possible.

As reported by the Ministry of New and Renewable Energy (MNRE), 5400 kilowatt-hour solar power was generated

and exported to the grid in a period of 90 days. The amount of power imported from the grid was 5300 kilowatt-hour. This indicates excess energy generation of 100 kilowatt-hour.

Another report illustrates that while Deshpriya Park's electricity bill for April 2014 was ₹ 31,000 before solar transition, it went down to as low as ₹ 1800 for April 2015 after solar installation.

**Reduced carbon footprint.** The limited dependence on grid electricity and transition to LED technology greatly reduced carbon

gas emission. According to Dr Gonchaudhuri, the setup cut carbon dioxide emission by 72 tonnes annually.

**No opex and no failure.** Battery-driven solar streetlights call for frequent maintenance and periodic replacement of batteries, usually within four years of use. This imposes a certain amount of operational expense throughout the product life. On the other hand, batteryless streetlights have no such requirements. Additionally, due to grid connectivity, these run on days with no sunlight as well. No disturbances were recorded in the streetlight functioning. Also, failure rate was negligible.

## Summing up

Based on Calcutta Electric Supply Corporation (CESC) tariffs, the return on investment is expected within four years of use. Carbon-neutral batteryless streetlights are great options for urban communities, especially flyovers, parks and many other public areas where electricity availability is consistent. Following the installation, Kolkata Municipal Corporation has taken up the target of solarising 28 other parks in the city, which will save them ₹ 2.5 million for each park annually. Similar initiatives across all metro-cities of India can bring a massive cumulative benefit to the nation. **EFY**

—Paromik Chakraborty, technical journalist, EFY

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